COMPASS Synoptic CB Porewater: DOC

September 2023 Samples

2026-06-03

##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

```
## Setup Information
```

```
###### Run information - PLEASE CHANGE
  Date_Run = "09/16/23"  #Date that instrument was run
  Run_by = "Stephanie J. Wilson"  #Instrument user
  Script_run_by = "Stephanie J. Wilson" #Code user
  run_notes = "NA"  #any notes from the run
  samples <- c("GCW", "GWI", "MSM", "SWH") #whatever identifies your samples within the same names
  samples_pattern <- paste(samples, collapse = "|")
    #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
  chks_name = "Chk_Std_50C_2N"  #what did you name your check standards?

###### File Names - PLEASE CHANGE
#file path and name for raw summary data file
    #raw_file_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505.txt" #example
    raw_file_name = c("Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202309_1.txt",
                      "Raw Data/TOCTN_COMPASS_Synoptic_TCTN_202309_2.txt")

#file path and name for raw all peaks file
    #raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202505_allpeaks.txt" #example
   # raw_allpeaks_name = "Raw Data/COMPASS_SynopticCB_PW_DOC_202411_allpeaks.txt"

#file path and name of processed data file
    #processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202505.csv" #example
    processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DOC_202309.csv" #example


###### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
  Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2023.csv"

#qaqc log file path for this year
  #Log_path = "Raw Data/COMPASS_Synoptic_TOCTN_QAQClog_2024.csv"
```

##Set Up Code

##Read in metadata and create similar sample IDs for matching to samples

Import Data Functions

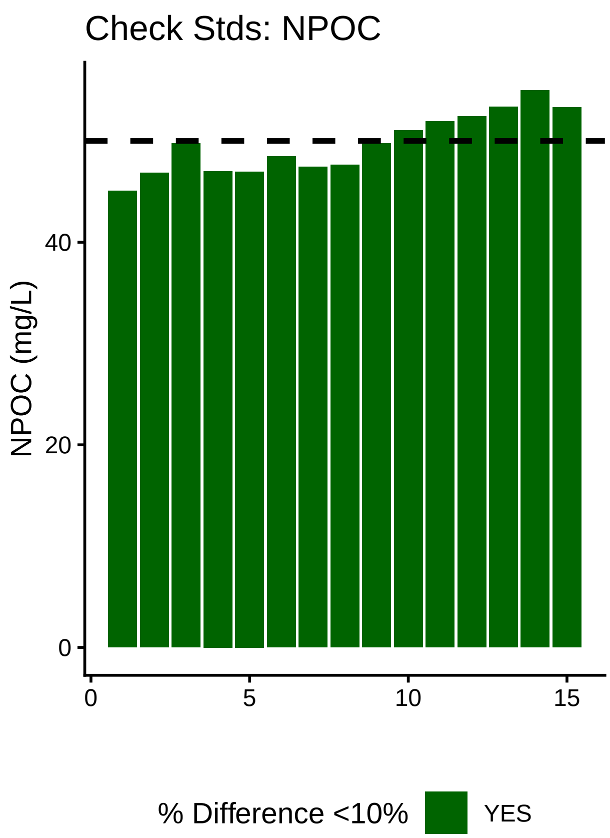
Import Sample Data

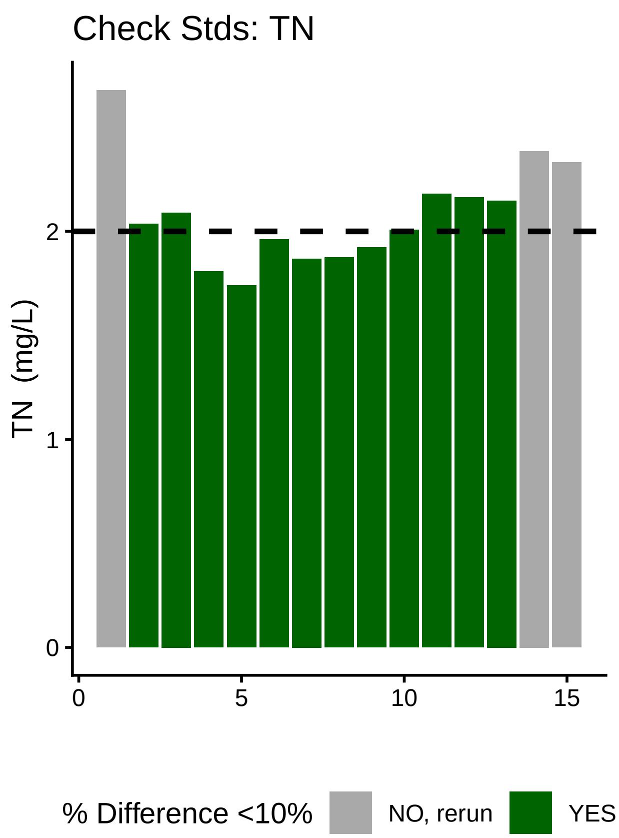
```
## Import Sample Data

## New names:
## New names:
## * `` -> `...14`

## # A tibble: 6 x 4
##   sample_name             npoc_raw tdn_raw run_datetime
##   <chr>                      <dbl>   <dbl> <chr>
## 1 202309_GCW_TR_LysA_20cm    35.8    1.42  9/16/2023 7:07:41 PM
## 2 202309_GCW_TR_LysA_45cm    13.9    0.527 9/16/2023 7:32:14 PM
## 3 202309_GCW_TR_LysB_10cm     4.85   0.393 9/16/2023 7:55:43 PM
## 4 202309_GCW_TR_LysB_20cm     3.60   2.85  9/16/2023 8:15:13 PM
## 5 202309_GCW_TR_LysB_45cm    21.1    0.776 9/16/2023 8:40:45 PM
## 6 202309_GCW_TR_LysC_10cm    13.0    0.336 9/16/2023 9:01:01 PM
```

Assessing Standard Curves - assessed manually on the instrument

Assess Check Standards

```
## Assess the Check Standards

## New names:
## New names:
## * `` -> `...14`

## [1] "Carbon Check Standard RSD within Range"

## [1] "Nitrogen CHECK STANDARD RSD TOO HIGH - REASSESS"
```

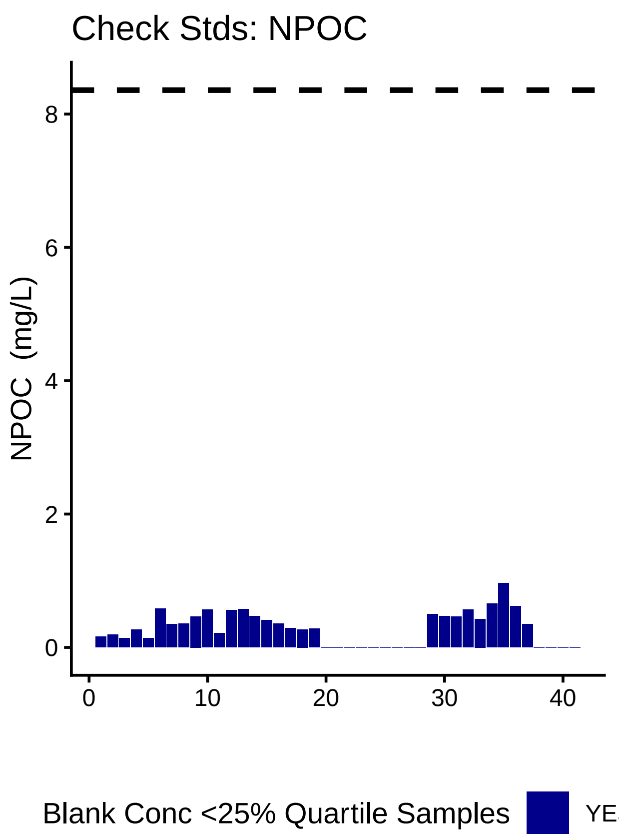


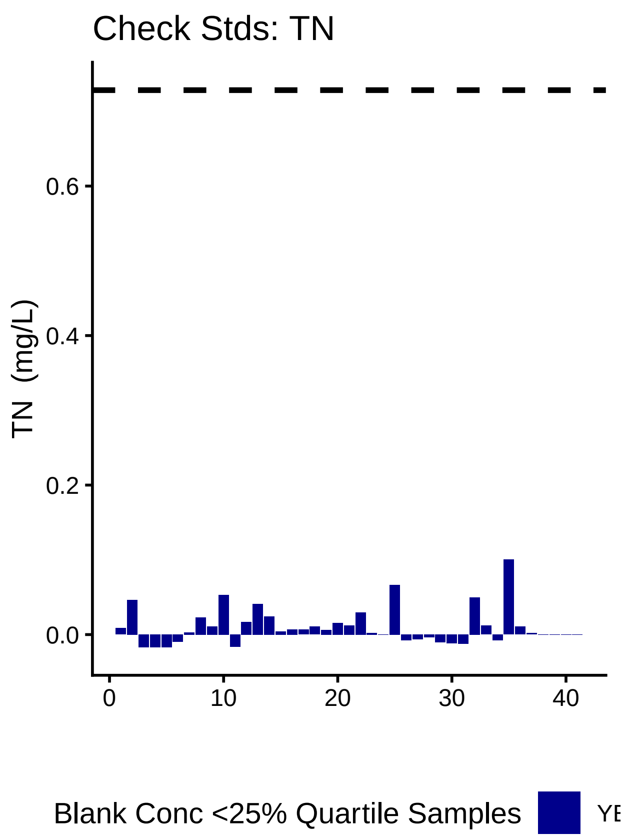



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"

## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

Assess Blanks

```
## Assess Blanks

## New names:
## New names:
## * `` -> `...14`

## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"

## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```

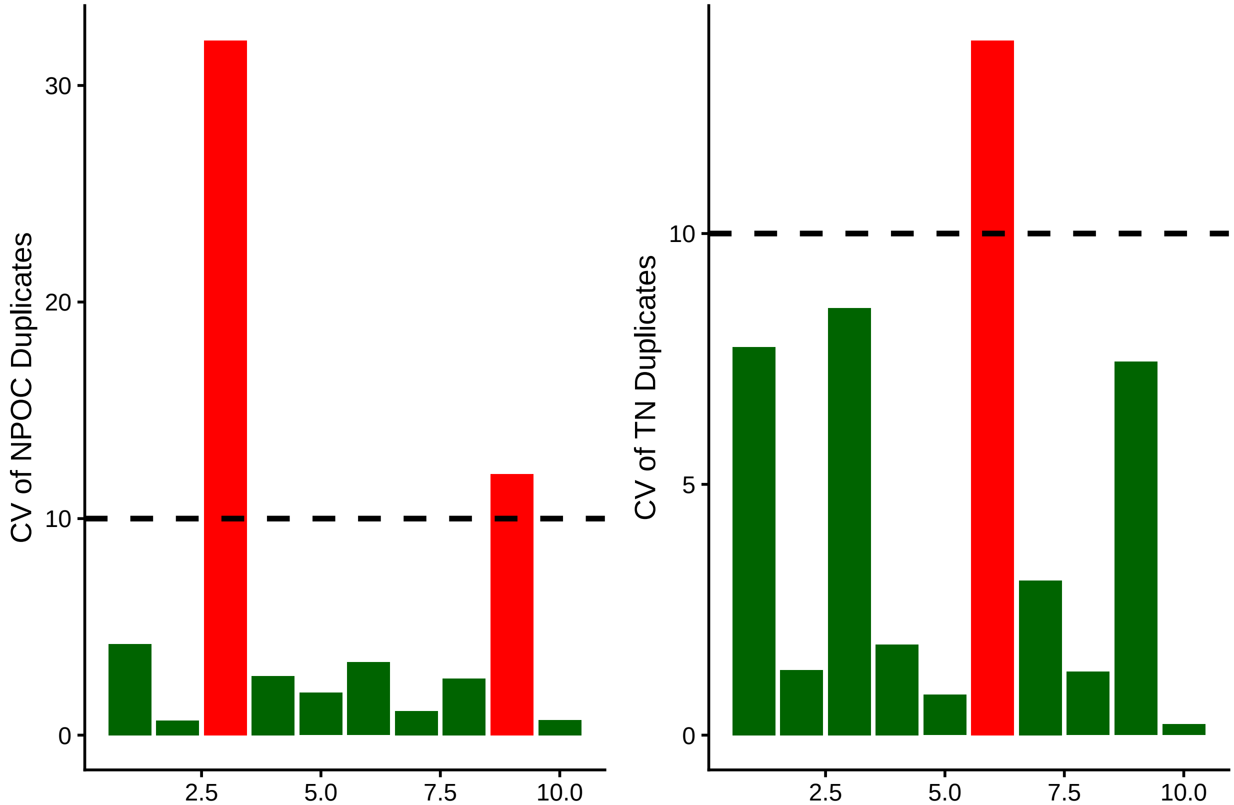





```
## carbon blanks:

## [1] 0.2844268

## nitrogen blanks:

## [1] 0.01043415
```

Assess Duplicates

```
## Assess Duplicates

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

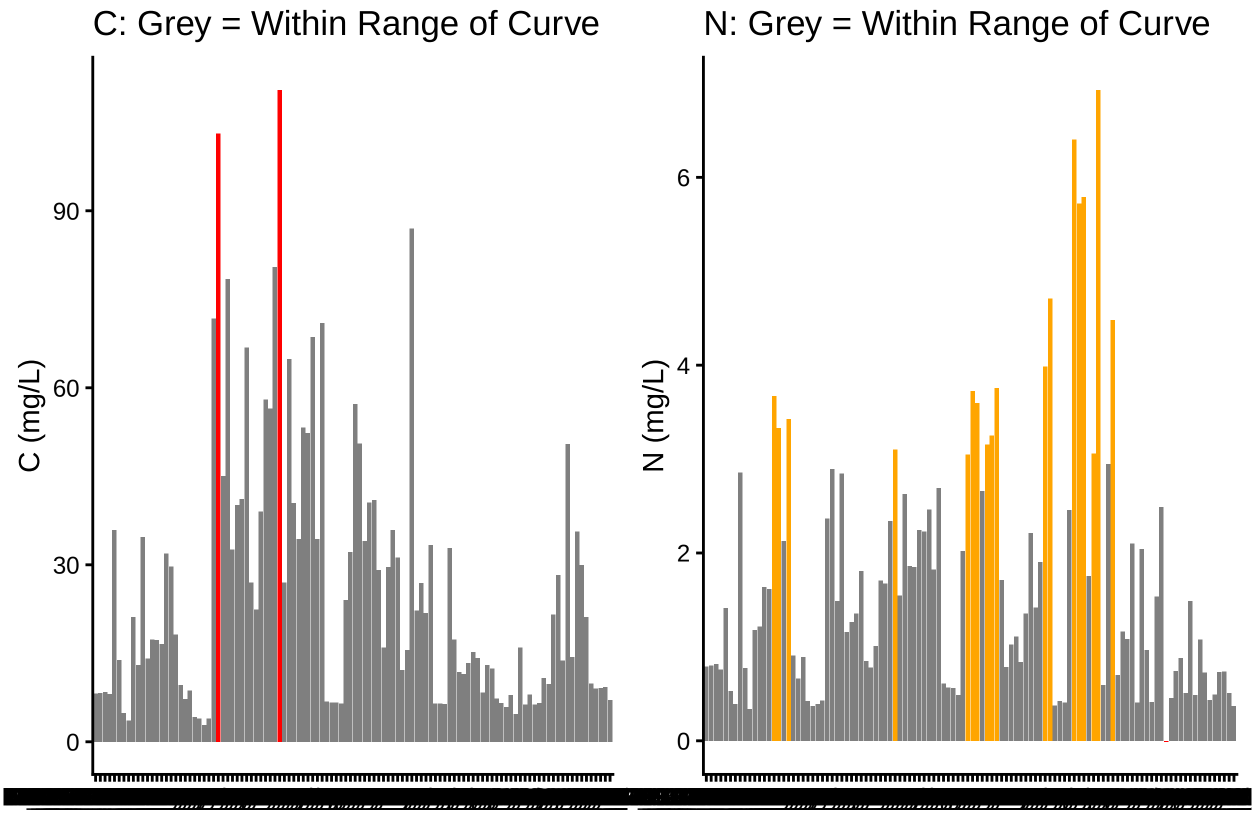



```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] ">60% of Nitrogen Duplicates have a CV <10%"
```

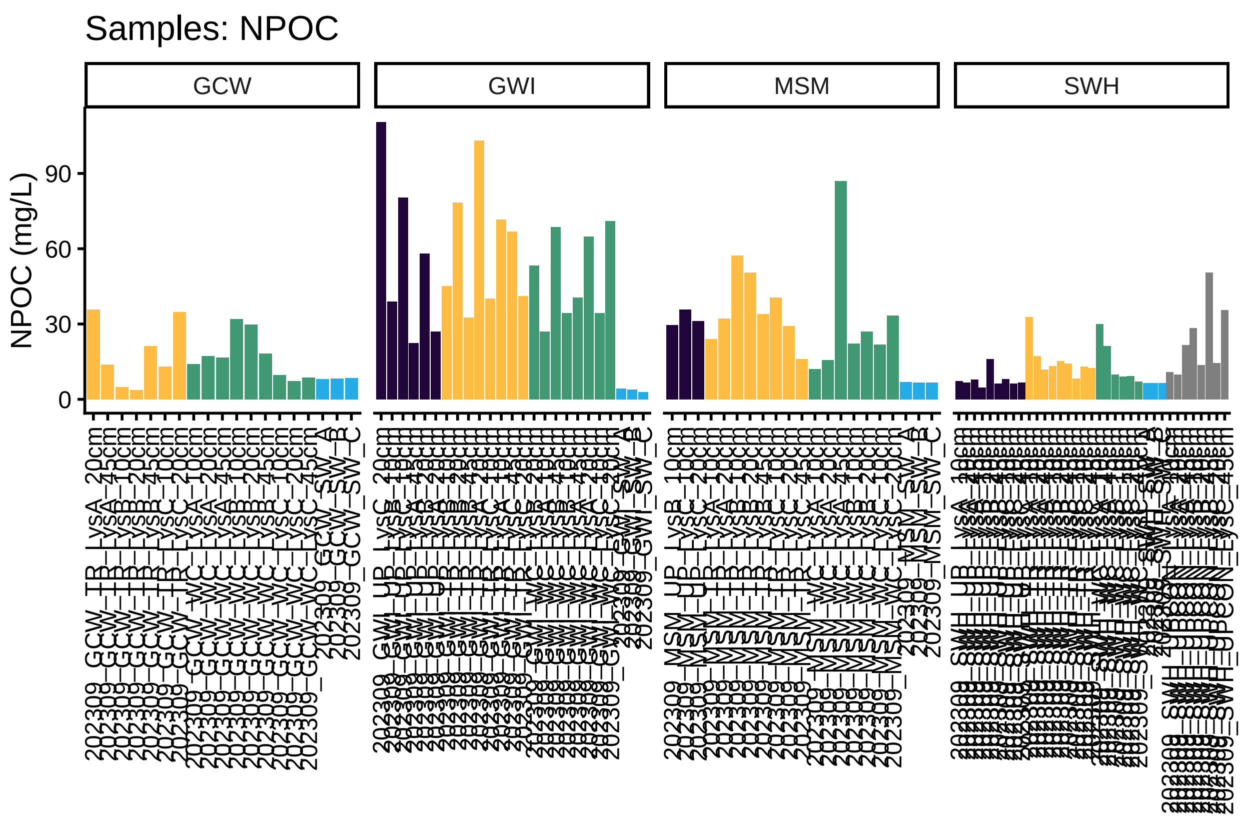
Sample Flagging - Are samples Within the range of the curve?

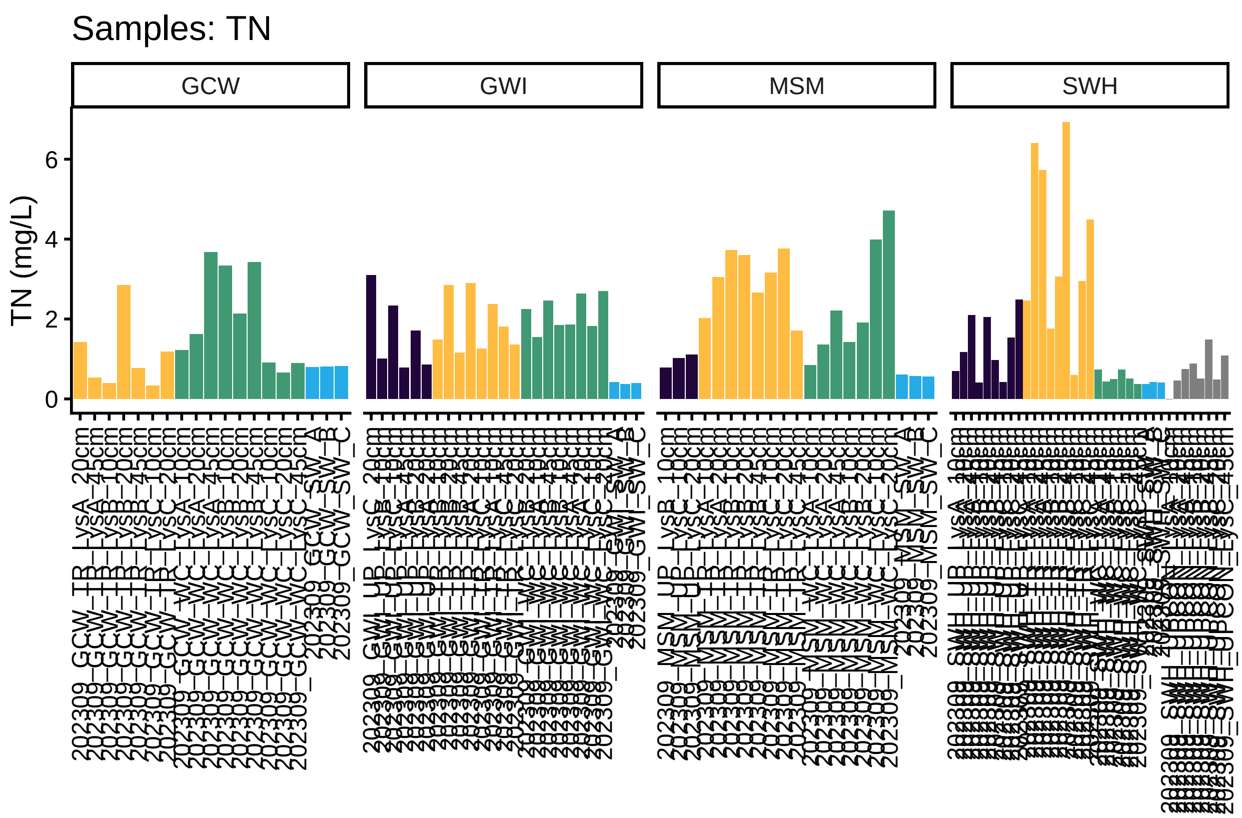
```
## Sample Flagging
```



Visualize Data by Plot

```
## Visualize Data

## Warning in rbind(c("202309", "GCW", "TR", "LysA", "20cm"), c("202309", "GCW", :
## number of columns of result is not a multiple of vector length (arg 17)

##  [1] Date          Site_Code     Zone          Lysimeter     Depth
##  [6] sample_name   npoc_raw      tdn_raw       run_datetime  npoc_flag
## [11] tdn_flag
## <0 rows> (or 0-length row.names)
```


Samples: NPOC
GCW
GWI
MSM
SWH
NPOC (mg/L)
90
60
30
0



Convert data from mg/L to uMoles/L

Check to see if samples run match metadata & merge info

```
## Check Sample IDs with Metadata

## Some sample IDs are missing from metadata.

## [1] "202309_MSM_UP_LysB_10cm" "202309_GWI_TR_LysC_45cm"
## [3] "202309_MSM_WC_LysA_45cm" "202309_MSM_WC_LysC_20cm"
## [5] "202309_SWH_WC_LysB_10cm" "202309_SWH_WC_LysC_10cm"
## [7] "202309_SWH_WC_LysC_20cm"
```

Export Processed Data

```
## Export Processed Data

## # A tibble: 6 x 21
##   Project      Region Site  Zone  Replicate Depth_cm Sample_ID  Year Month   Day
##   <chr>        <chr>  <chr> <fct> <chr>        <int> <chr>     <int> <int> <int>
## 1 COMPASS: Sy~ CB     MSM   UP    <NA>            NA 202309_M~    NA    NA    NA
## 2 COMPASS: Sy~ CB     MSM   UP    C               10 202309_M~  2023     9    14
## 3 COMPASS: Sy~ CB     MSM   UP    C               20 202309_M~  2023     9    14
```

```
## 4 COMPASS: Sy~ CB     SWH   UP    A               10 202309_S~  2023     9    20
## 5 COMPASS: Sy~ CB     SWH   UP    A               20 202309_S~  2023     9    20
## 6 COMPASS: Sy~ CB     SWH   UP    A               45 202309_S~  2023     9    20
## # i 11 more variables: Time <lgl>, Time_Zone <lgl>, npoc_mgL <dbl>,
## #   npoc_uM <dbl>, npoc_flag <chr>, tdn_mgL <dbl>, tdn_uM <dbl>,
## #   tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>, Field_notes <chr>
```

#end